AtliQ Hardware – Ad-hoc Consumer Analysis

SQL + Power BI Project

Objectives

- AtliQ Hardware is one of the major computer hardware manufacturers in India, with a strong presence in other nations.
- Nevertheless, the management did note that they do not have sufficient insights to make prompt, wise, and data-informed judgments.
- So, they plan to expand the data analytics team by adding junior data analysts.
- To assess candidates, Data analytics director, Tony Sharma plans to conduct a SQL challenge to evaluate both tech and soft skills.
- The company seeks insights for 10 ad hoc requests.

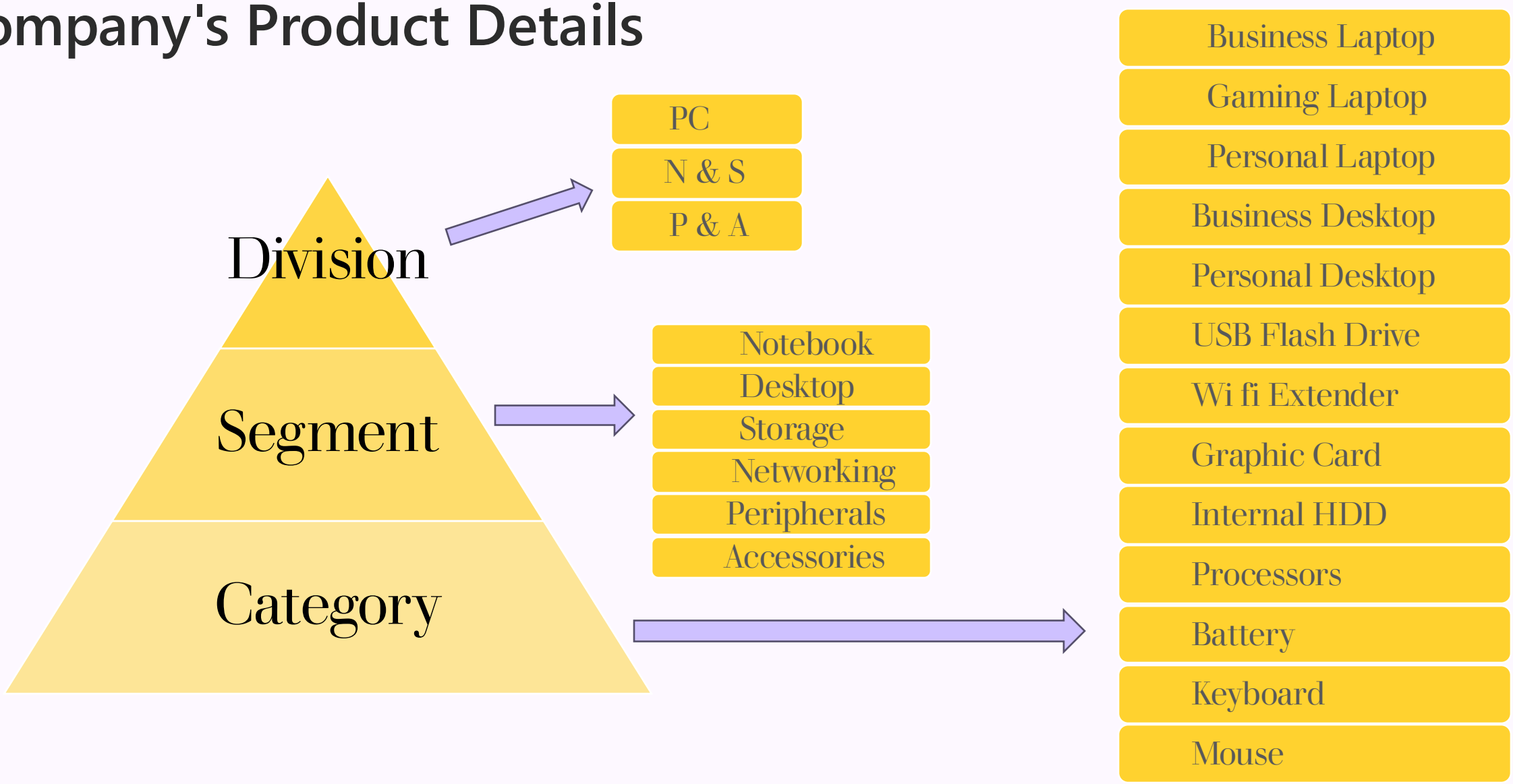
Company Details

AtliQ Hardware is a computer hardware and accessory manufacturer.

FY 2020
SEPT 2019
- AUG
2020

FY 2021
SEPT 2020
- AUG
2021

Company's Product Details



Company's Market

NA

- Cannada
- USA

LATAM

- Brazil
- Chile
- Columbia
- Mexico

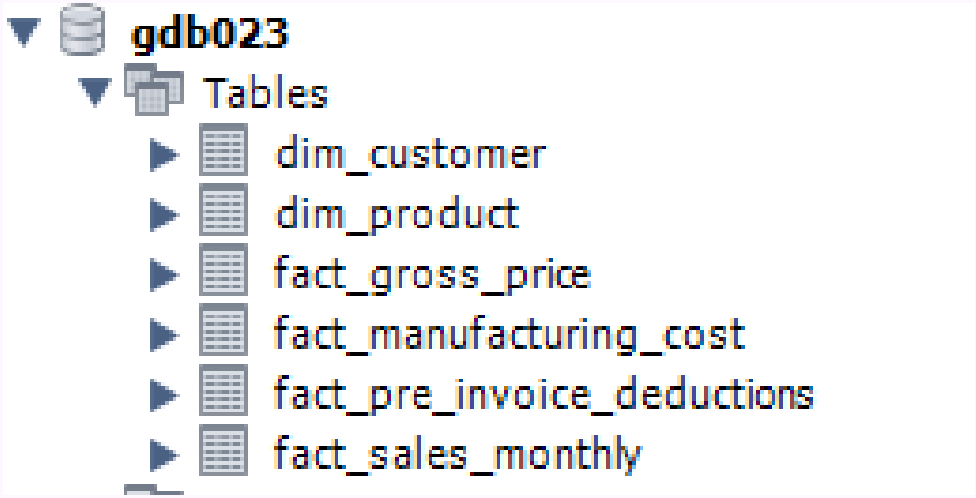
EU

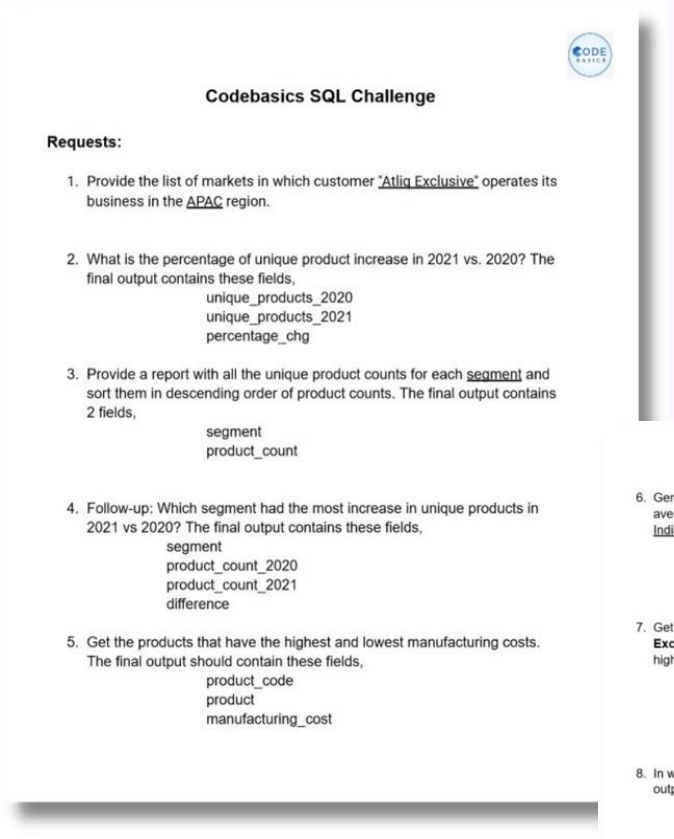
- Austria
- Italy
- Spain
- Portugal
- France
- Netherlands
- Germany
- Poland
- United Kingdom
- Norway
- Sweden

APAC

- Australia
- India
- New Zealand
- China
- Japan
- Indonesia
- Philippines
- South Korea
- Bangladesh
- Pakistan

Data, Requests, and Tools





Codebasics SQL Challenge

Requests:

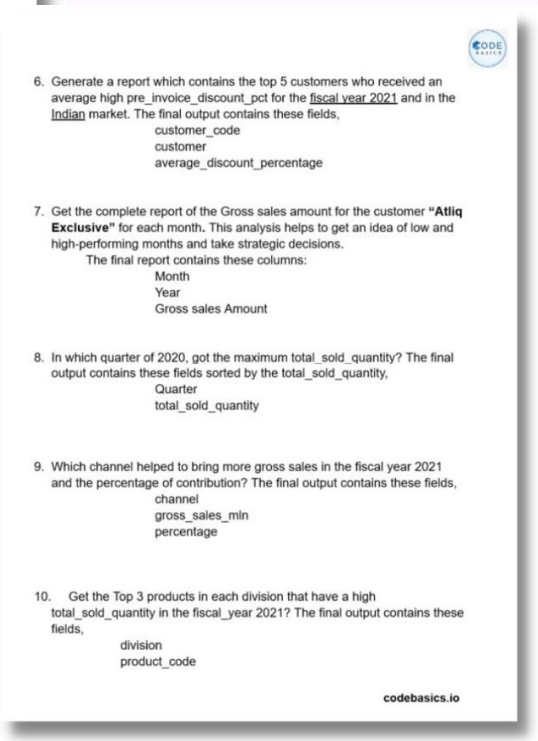
1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,
   unique_products_2020
   unique_products_2021
   percentage_chg

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,
   segment
   product_count

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,
   segment
   product_count_2020
   product_count_2021
   difference

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,
   product_code
   product
   manufacturing_cost



6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
   customer_code
   customer
   average_discount_percentage

7. Get the complete report of the Gross sales amount for the customer **"Atliq Exclusive"** for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.
   The final report contains these columns:
   Month
   Year
   Gross sales Amount

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,
   Quarter
   total_sold_quantity

9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,
   channel
   gross_sales_mln
   percentage

10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,
    division
    product_code

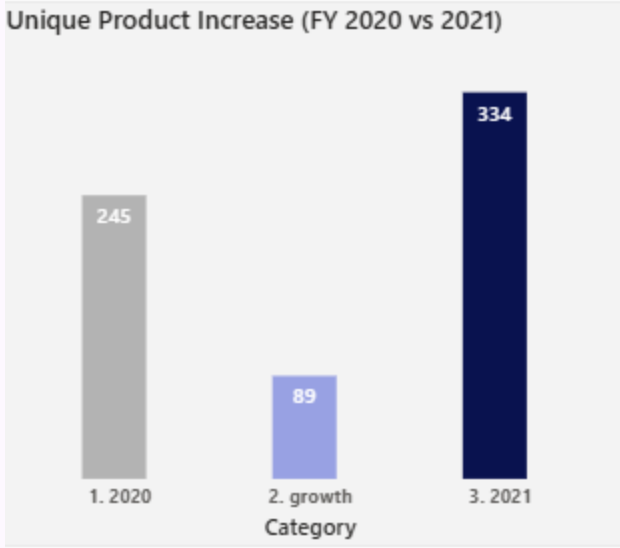
codebasics.io

1. Provide the list of markets in which customer "AtliQ Exclusive" operates its business in the APAC region.

market
Australia
Bangladesh
India
Indonesia
Japan
Newzealand
Philiphines
South Korea

2. What is the percentage of unique product increase in 2021 vs. 2020?
The final output contains these fields,
unique_products_2020
unique_products_2021
percentage_chg

unique_product_2020	unique_products_2021	percentage_chg
245	334	36.33

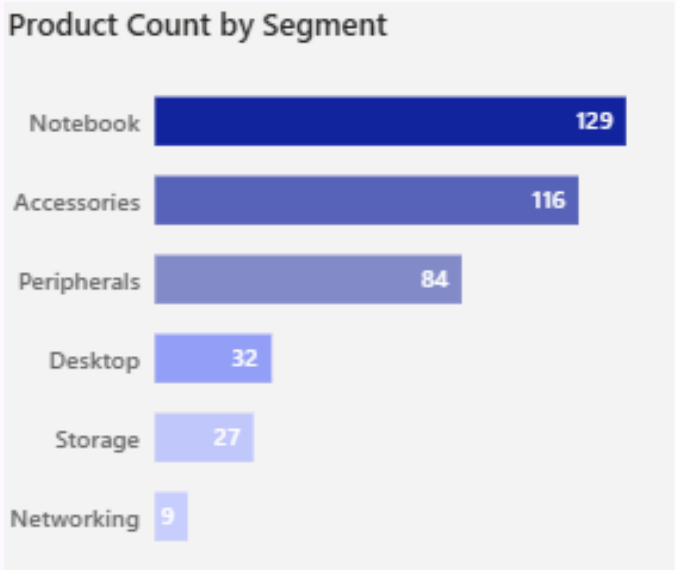


INSIGHT :

The number of unique products increased from **245 in 2020 to 334 in 2021**, reflecting a **36.33% growth** in the product portfolio. This indicates a strong expansion strategy, likely driven by increased market demand and a focus on diversification.

3. Provide a report with all the unique product counts for each segment and sort them in descending order of
product counts. The final output contains 2 fields,
segment
product_count

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

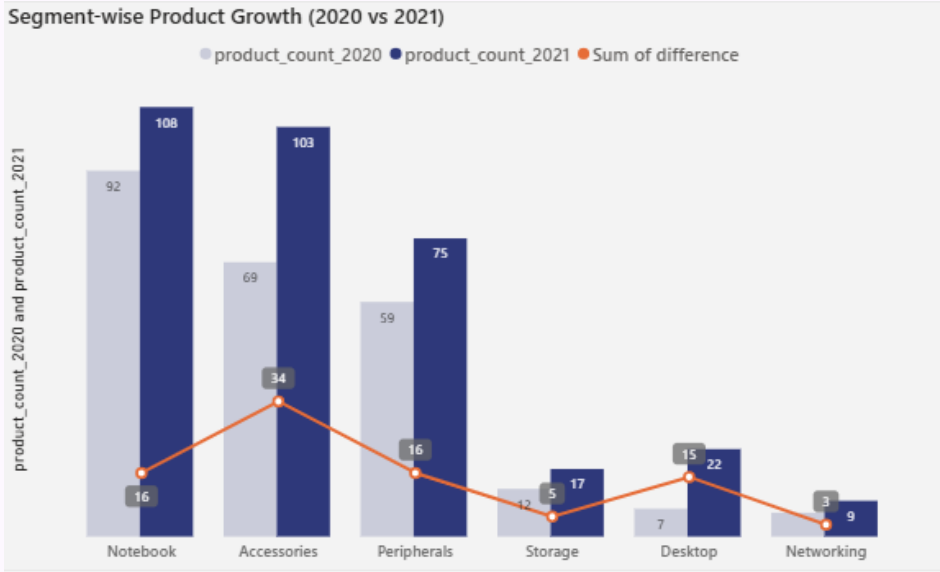


INSIGHT :

We can see that the Notebook and Accessories segments dominate the product portfolio, while segments like Networking and Storage have very few products. This suggests that the company is focusing more on certain high-demand categories, but there may be opportunities to expand in underrepresented segments.

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,
segment
product_count_2020
product_count_2021
difference

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Desktop	7	22	15
Networking	6	9	3
Notebook	92	108	16
Peripherals	59	75	16
Storage	12	17	5



INSIGHT :

Accessories:

- Strong expansion → high demand segment

Networking:

- Low growth → potential saturation or low focus

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,
product_code
product
manufacturing_cost

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Products with Highest & Lowest Manufacturing Cost

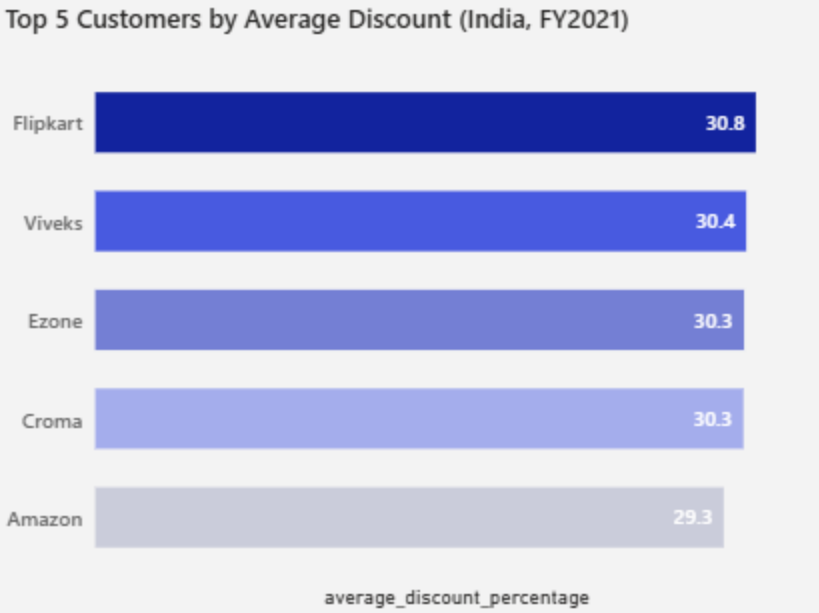
product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.54
A2118150101	AQ Master wired x1 Ms	0.89

INSIGHT :

- High-cost:
  - Needs strong pricing strategy
- Low-cost:
  - Opportunity for high margins

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
customer_code
customer
average_discount_percentage

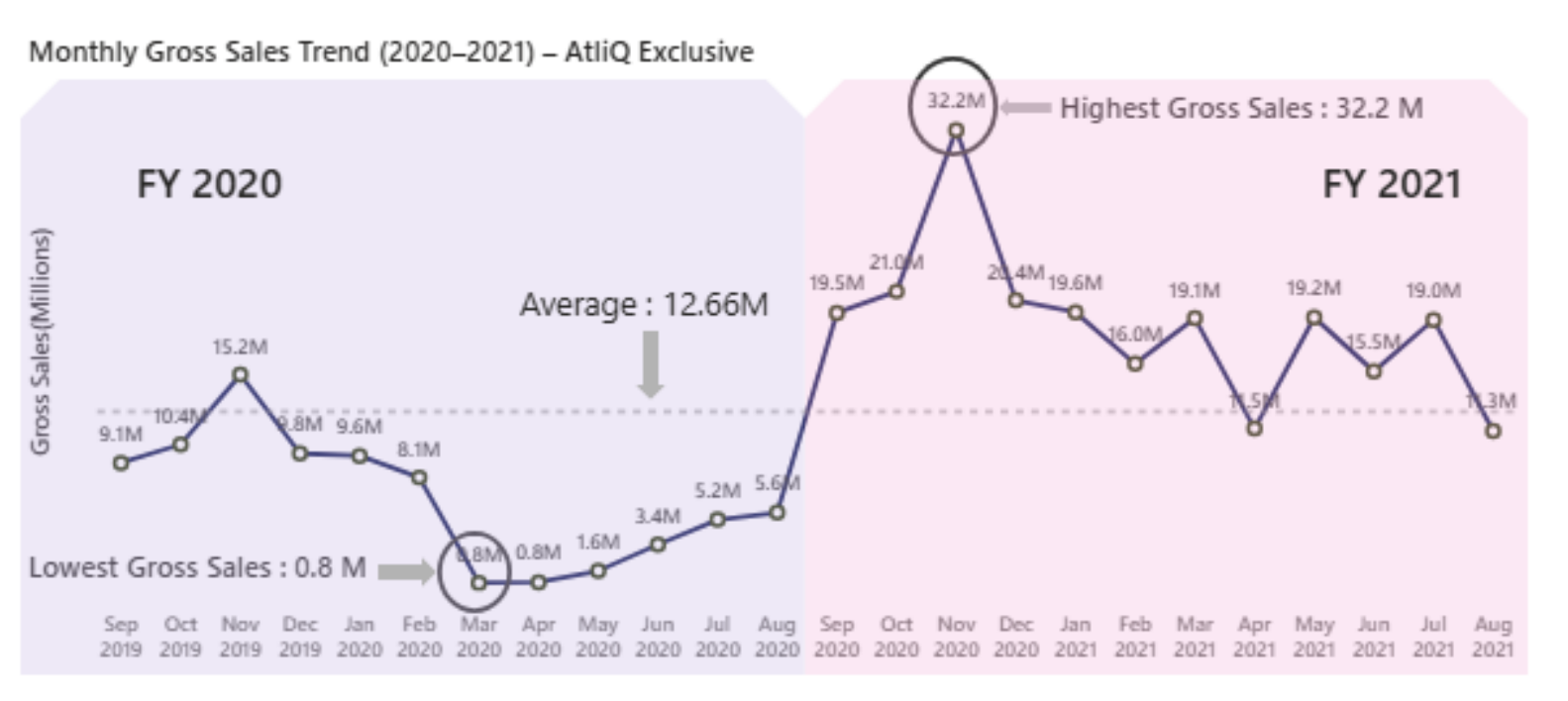
customer_code	customer	average_discount_percentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933



INSIGHT :

A small group of customers receives significantly higher discounts, indicating strong negotiation power or strategic partnerships.

7. Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns:
Month
Year
Gross sales Amount

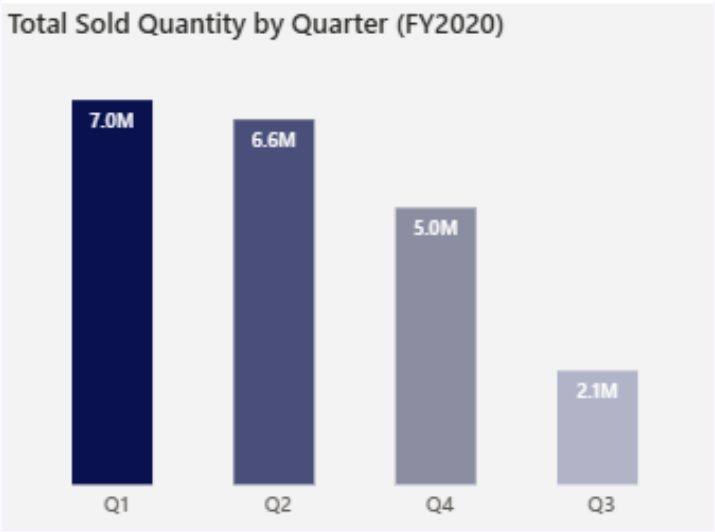


INSIGHT :

From the trend, we can see that sales in **FY2021** are significantly higher and more stable compared to **FY2020**. The average sales have increased, and the company achieved its hi ghest monthly sales of around **32 million**, indicating strong growth and recovery.

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,
Quarter
total_sold_quantity

Quarters	total_sold_quantity
1	7005619
2	6649642
4	5042541
3	2075087

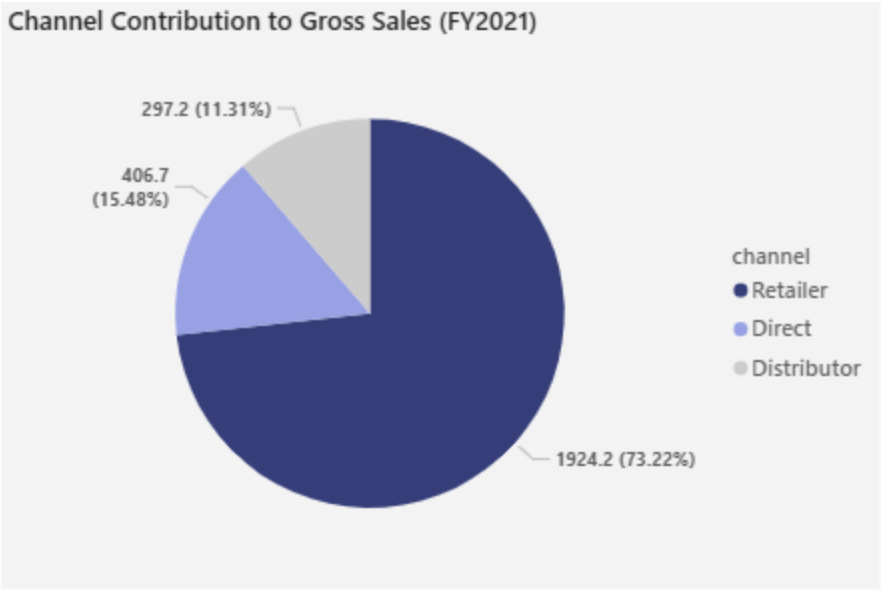


INSIGHT :

We can see that **Q1** had the highest total sold quantity, around **7 million** units, while **Q3** recorded the lowest. This indicates that demand is strongest at the beginning of the fiscal year and dips in the middle, which can help in planning inventory and marketing strategies

9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,
channel
gross_sales_mln
percentage

channel	Gross_sales_mln	percentage
Retailer	1924.17 M	73.22 %
Direct	406.69 M	15.48 %
Distributor	297.18 M	11.31 %



INSIGHT :

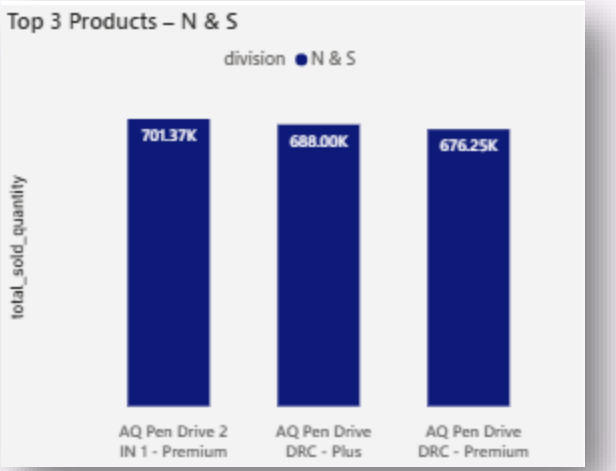
The **Retailer channel dominates gross sales in FY2021, contributing approximately 73%** of total revenue. In comparison, **Direct (15%)** and **Distributor (11%)** channels contribute significantly less, indicating a heavy reliance on the Retailer channel.

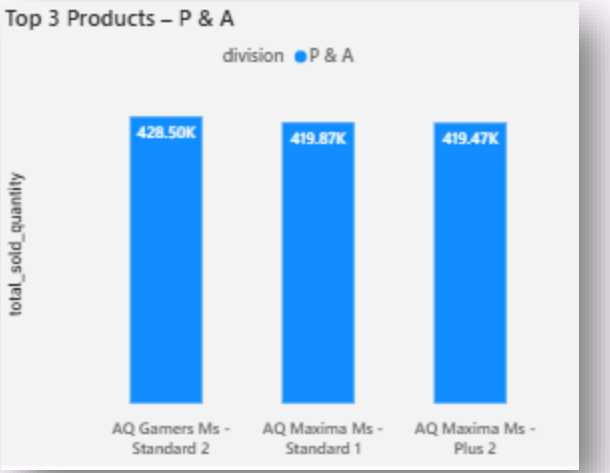
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields, division
product_code, product
total_sold_quantity
rank_order

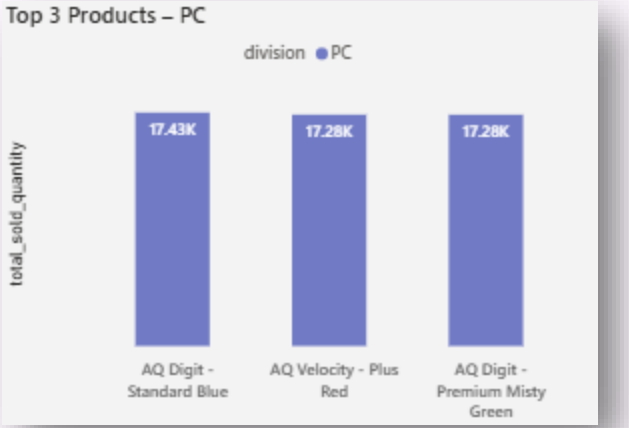
division	product_code	product	Total_sold_quantity	Rank_Order
N & S	A6720160103	AQ Pen Drive 2 IN 1 [Premium]	701373	1
N & S	A6818160202	AQ Pen Drive DRC [Plus]	688003	2
N & S	A6819160203	AQ Pen Drive DRC [Premium]	676245	3
P & A	A2319150302	AQ Gamers Ms [Standard 2]	428498	1
P & A	A2520150501	AQ Maxima Ms [Standard 1]	419865	2
P & A	A2520150504	AQ Maxima Ms [Plus 2]	419471	3
PC	A4218110202	AQ Digit [Standard Blue]	17434	1
PC	A4319110306	AQ Velocity [Plus Red]	17280	2
PC	A4218110208	AQ Digit [Premium Misty Green]	17275	3

INSIGHT :

Each division has distinct top-performing products, with the N & S division leading significantly in sales volume, followed by P & A, while the PC division shows comparatively lower performance, highlighting differences in demand across product categories.


Top 3 Products – N & S
division N & S
total_sold_quantity
701.37K
688.00K
676.25K
AQ Pen Drive 2 IN 1 - Premium
AQ Pen Drive DRC - Plus
AQ Pen Drive DRC - Premium


Top 3 Products – P & A
division P & A
total_sold_quantity
428.50K
419.87K
419.47K
AQ Gamers Ms - Standard 2
AQ Maxima Ms - Standard 1
AQ Maxima Ms - Plus 2


Top 3 Products – PC
division PC
total_sold_quantity
17.43K
17.28K
17.28K
AQ Digit - Standard Blue
AQ Velocity - Plus Red
AQ Digit - Premium Misty Green

Thank You